Ming Tai

Curriculum Vitae

PhD Student, Renmin University of China
Database Systems

Education

2026–Present
Renmin University of China
PhD student studying database systems.

2022–2026
Chengdu University of Technology
Bachelor's degree in Intelligent Science and Technology. Ranked first of 123 students in July 2025.

Research projects

Current research
MAVRA
Built a shared memory prototype for data agents in Rust and PostgreSQL. It tracks the validity of metric definitions and checks revisions at query time.

Jan–Jun 2025
Transaction routing in KingbaseRAC
Implemented a C++ router in a collaboration between RUC and Kingbase. It uses METIS to group transactions by access pattern and limit cross-node traffic.

Oct 2024–Jan 2025
OceanBase Database Competition
Led the RushDB team. We implemented nested and correlated subqueries in MiniOB and optimized vector search in OceanBase's VSAG library.

2024
RucBase database system
Implemented B+ tree indexes, locking for concurrent transactions, and crash recovery using write-ahead logging and checkpoints.

Internships

Started Jul 2024
OceanBase, Ant Group
Database research intern during undergraduate study.

Jan–Feb 2024
Chengdu Yinguang Software
Developed front-end and back-end components of an operations management system using Vue 3 and Spring Boot.

Honors

2025
National Scholarship
Undergraduate award, Chengdu University of Technology.

2024 and 2025
National First Prize, Database Management System Design
National College Computer System Competition. RushDB placed second overall in 2025.

Jan 2025
Fourth OceanBase Database Competition
RushDB placed third overall and first among undergraduate teams in the final.

Undergraduate
Other competition awards
National First Prize, Fourth International Supply Chain Modeling Competition. Provincial First Prize, 14th Lanqiao Cup Programming Competition.